



Faculty of Computing & Information Technology

CC-215-L: Database Systems Lab

BS(SE) Morning - Fall 2023, Semester Spring 2025

Time Allowed: 90 Minutes

LAB – 09

[55 Marks]

Task 1: Create Table [15 Marks]

Create the following tables for a Sales System. You must give meaningful names to all the constraints.

Table Name: customer		
Col Name	Data Type	Constraints
cust#	NUMBER(5)	Primary Key
c_name	VARCHAR2(20)	Not Null
city	VARCHAR2(30)	
cnic	CHAR(16)	Unique

Table Name: invoice		
Col Name	Data Type	Constraints
invoice#	NUMBER(5)	Primary Key
inv_date	DATE	DEFAULT sysdate
cust#	NUMBER(5)	Foreign Key references Customer
payment	CHAR(12)	Cash, Cheque, Credit Card
comments	VARCHAR2(50)	

Table Name: product		
Col Name	Data Type	Constraints
prod#	NUMBER(5)	Primary Key
p_name	VARCHAR2(50)	Not Null
price	NUMBER(5)	Above 5

Table Name: invoice_details		
Col Name	Data Type	Constraints
invoice#	NUMBER(5)	Primary Key, Foreign Key references Invoice
prod#	NUMBER(5)	Primary Key
qty_ordered	NUMBER(4)	
sale_price	NUMBER(5)	Above 5

Task 2:**Alter Table****[15 Marks]**

Perform the following tasks on the above relational schema.

1. Add the following attributes in the product table:

Col Name	Data Type	Constraints
stock	NUMBER(5)	Must not be negative

2. Change the table name **invoice** to **sales_invoice**.
3. Change the column name price (of product) to cost_price.
4. Change the column names (of sales_invoice) invoice# and inv_date to s_invoice# and s_inv_date respectively of
5. Change the table name **invoice_details** to **sales_invoice_details**.
6. Change the column names (of sales_invoice_details) invoice# to s_invoice#.
7. Apply FOREIGN KEY constraint on prod# of **sales_invoice_details**.
8. Apply NOT NULL constraint on cust# of **sales_invoice**.
9. Apply CHECK constraint on qty_ordered of **sales_invoice_details**, so that qty_ordered ranges between 1 and 500.
10. Drop comments column of **sales_invoice** table.

Task 3:**Insert Statement****[08 Marks]**

1. Populate the above tables with the following data.

Customer

cust#	c_name	city	cnic
49273	Ali	Karachi	35202-5698632-1
86741	Fatima	Lahore	35202-4659876-5
21598	Ahmed	Islamabad	35202-4987320-1
73956	Khadija	Rawalpindi	56987-1727537-3
60482	Hassan	Faisalabad	42569-2369715-7

Product

prod#	p_name	cost_price	stock
86741	Wireless Keyboard	2500	200
21598	Leather Wallet	1500	50
73956	LED Bulb	500	70
60482	Smart Watch	3500	40
16532	Bluetooth Earphones	1200	150

sales_invoice

s_invoice	s_inv_date	cust#	payment
101	May 3, 2025	49273	Cash
102	May 3, 2025	86741	Cheque
103	May 4, 2025	21598	Credit card

Sales_invoice_details

s_invoice#	prod#	qty_ordered	sale_price
101	21598	1	1600
101	73956	1	550
102	16532	5	1250
102	60482	1	3600
102	86741	2	2500
103	86741	2	2600
103	60482	1	3600
103	21598	1	1600
103	73956	1	550
103	16532	1	1250

Task 4: [12 Marks]

1. Create a table sales_invoice_copy using sales_invoice.
2. Create an empty table client using customer table.
3. Find the name of the most expensive product.
4. Find the name of most sold product.
5. Write a query to display s_invoice#, s_inv_date, s_inv_total, and c_name. The attribute s_inv_total is a derived attribute.
6. Find the names of the all the customers who have purchased a certain product.
7. Find the date(s) at which the highest invoices were generated.

Task 5: [05 Marks]

Viva Voce

Task 6: [15 Marks]

Post Lab

1. Create the following tables:

Table Name: vendor		
Col Name	Data Type	constraints
v_code	CHAR(5)	Primary Key
v_name	VARCHAR2(20)	Not Null
v_addr	VARCHAR2(30)	

Table Name: purchase_invoice		
Col Name	Data Type	constraints
p_invoice#	NUMBER(5)	Primary Key
p_invoice date	DATE	DEFAULT sysdate
v_code	NUMBER(5)	Foreign Key references vendor Not Null

Table Name: purchase_invoice_details		
Col Name	Data Type	Constraints
p_invoice#	NUMBER(5)	Primary Key, Foreign Key references purchase_invoice
prod#	NUMBER(5)	Primary Key
qty_purchased	NUMBER(4)	Must not be negative

2. Populate the above tables with the following data.

VENDOR

V_CODE	V_NAME	V_ADDR
V001	ALI	1234 Main Market
V002	HAMZA	5678 Johar Town
V003	HASSAN	91011 Garden Town

PURCHASE_INVOICE

p_invoice	p_inv_date	v_code
101	May 2, 2025	V001
102	May 2, 2025	V001
103	May 2, 2025	V003

PURCHASE_INVOICE_DETAILS

p_invoice#	prod#	qty_purchased
10001	21598	10
10001	73956	11
10002	16532	50
10002	60482	15
10002	86741	22
10003	86741	25
10003	60482	16
10003	21598	13
10003	73956	10
10003	16532	10